



Lecture Session 19

1. Evolution;
2. Classification of marine organisms;
3. Food web;
4. Factors impacting life in the ocean;

An aerial photograph of a coral reef system. The top left shows deep blue ocean water. A curved reef edge runs diagonally across the middle, featuring a sandy beach and a shallow lagoon with visible coral patterns. The bottom right shows a vast, shallow area with a greenish-brown seabed. The text "Life in the ocean" is centered in white.

Life in the ocean



Eye of Science/Science Source

Life on Earth – Fundamentally the same, but packaged differently

- Diverse, Diversity – different kinds of living organisms
- Unity – all species share the same underlying mechanisms for capturing and storing energy, manufacturing proteins, and transmitting information between generations

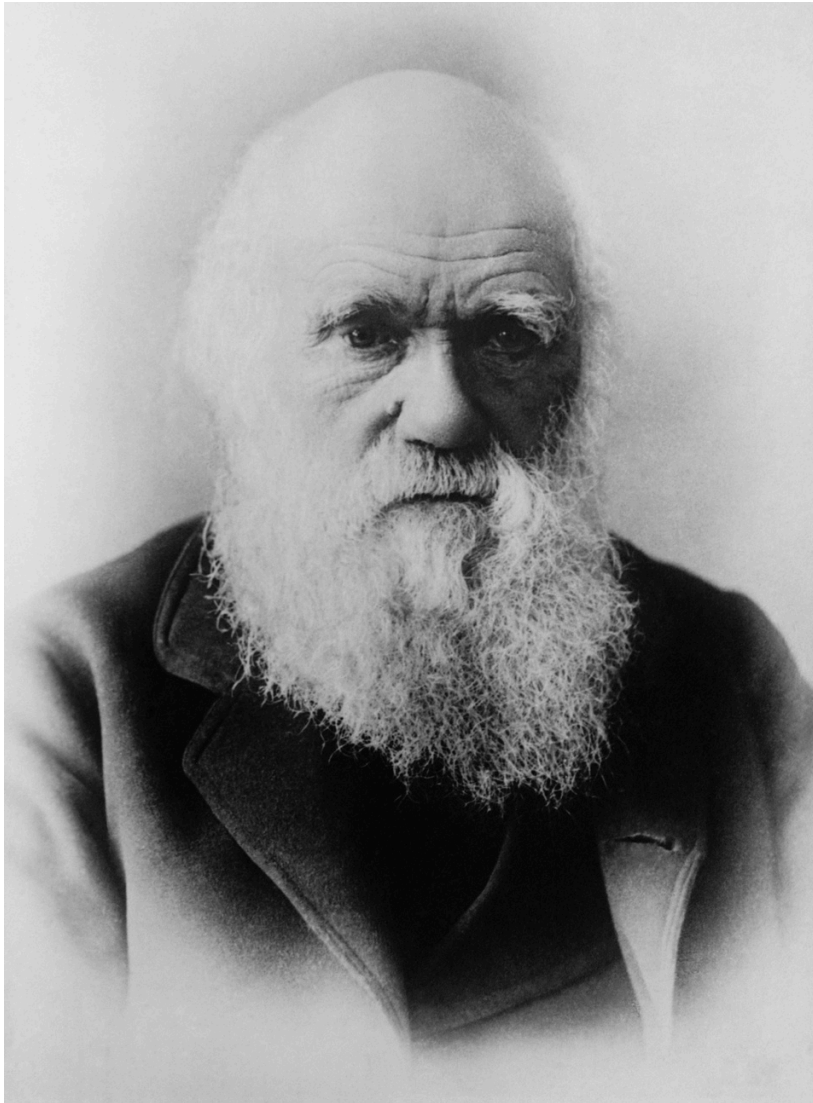
Figure 13.1 A tardigrade, a small aquatic animal with eight legs. Commonly called “water bears,” these tiny (1 millimeter; 0.04 inch) animals are noted for their ability to survive extreme conditions of temperature, pressure, ionizing radiation, starvation, and even complete vacuum. About a million tardigrades exist for every one of us.

Evolution

- Means change. It is the ability of living beings to change through time to fit the physical and chemical environment

Individuals do not "evolve" (biologically speaking), evolution takes generations.
On Earth, life had ~4 billions of years to evolve

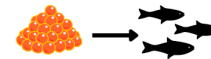




Natural Selection

Process through which living beings evolve

1- More offspring are produced than can survive to reproductive age



2- Random variations (mutations) occur in all organisms. Some of these traits are inheritable.



3- Some traits make an organism better suited to its environment



4- Favorable traits tend to accumulate in a population because organisms with favorable traits tend to survive more than those with unfavorable traits. They are selected.

5- The physical and biological environment affect the traits that are passed down from one generation to the next. favorable traits will show up more often in succeeding generations.

One World Ocean: relative "uniform" conditions of the oceans, makes its' organisms prone to convergent evolution

Two species that do not share a common ancestor have similar features. This is because the environment in which they evolved were similar.

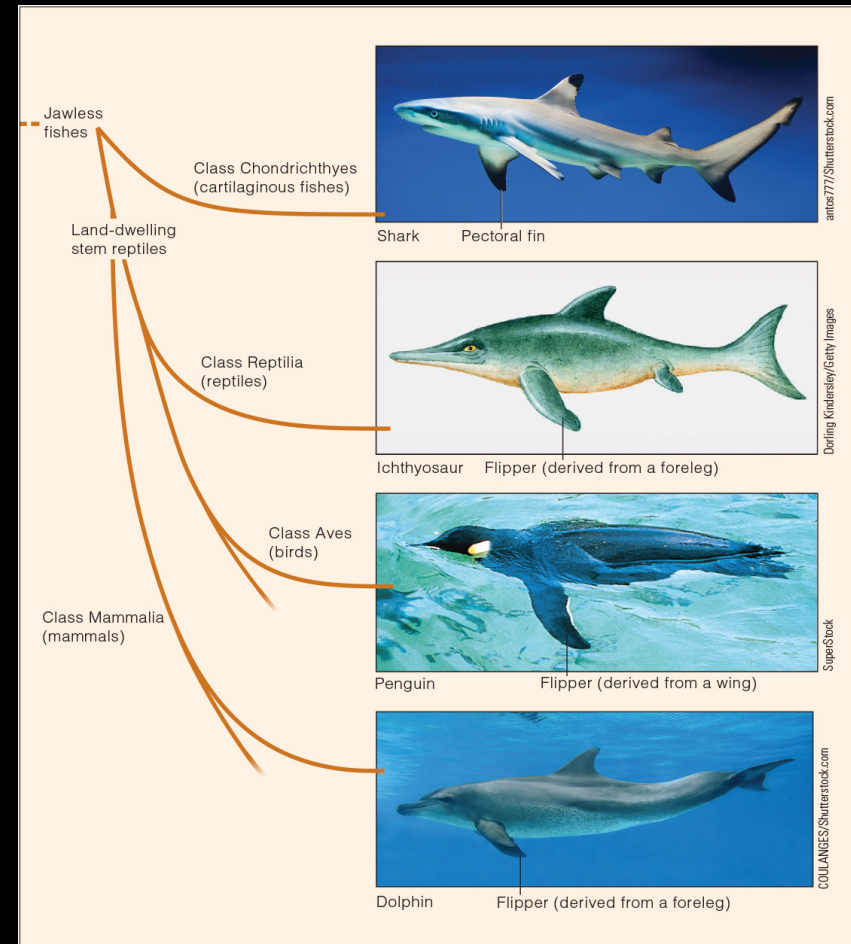


Figure 13.4 Convergent evolution in sharks, ichthyosaurs, penguins, and dolphins. Selection for adaptations that permitted rapid swimming resulted in superficially similar shapes among these four kinds of vertebrates, even though they are only remotely related.

An aerial photograph of a coral reef system. The top half of the image shows deep blue ocean water. Below this is a vibrant, multi-colored reef edge with shades of green, yellow, and orange. Further down is a wide, light-colored sandy beach. The bottom half of the image shows a vast, shallow lagoon with a greenish-brown hue, possibly due to algae or sediment. The title 'Classification of Life' is centered over the reef edge in a large, white, sans-serif font.

Classification of Life

Classification is done by evolutionary heritage

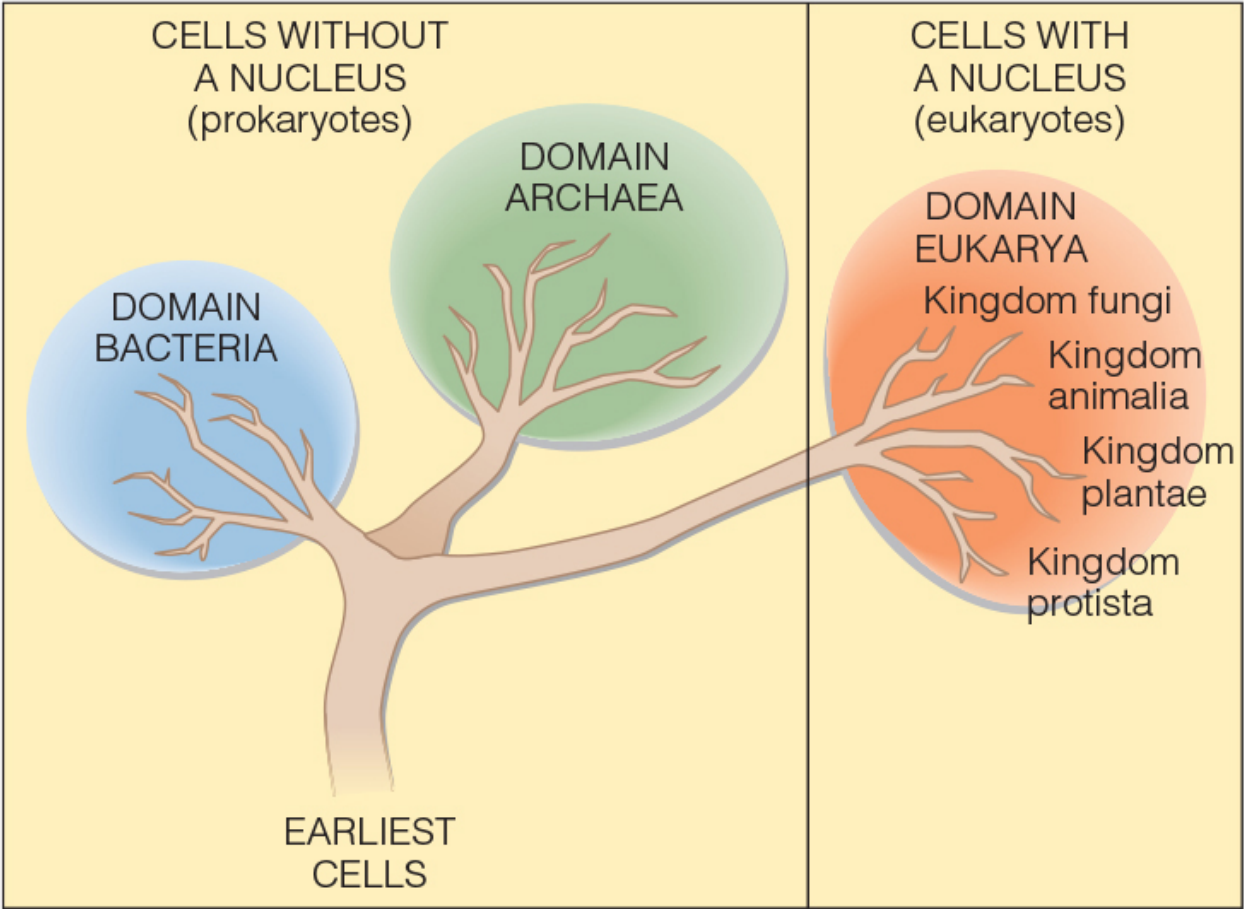
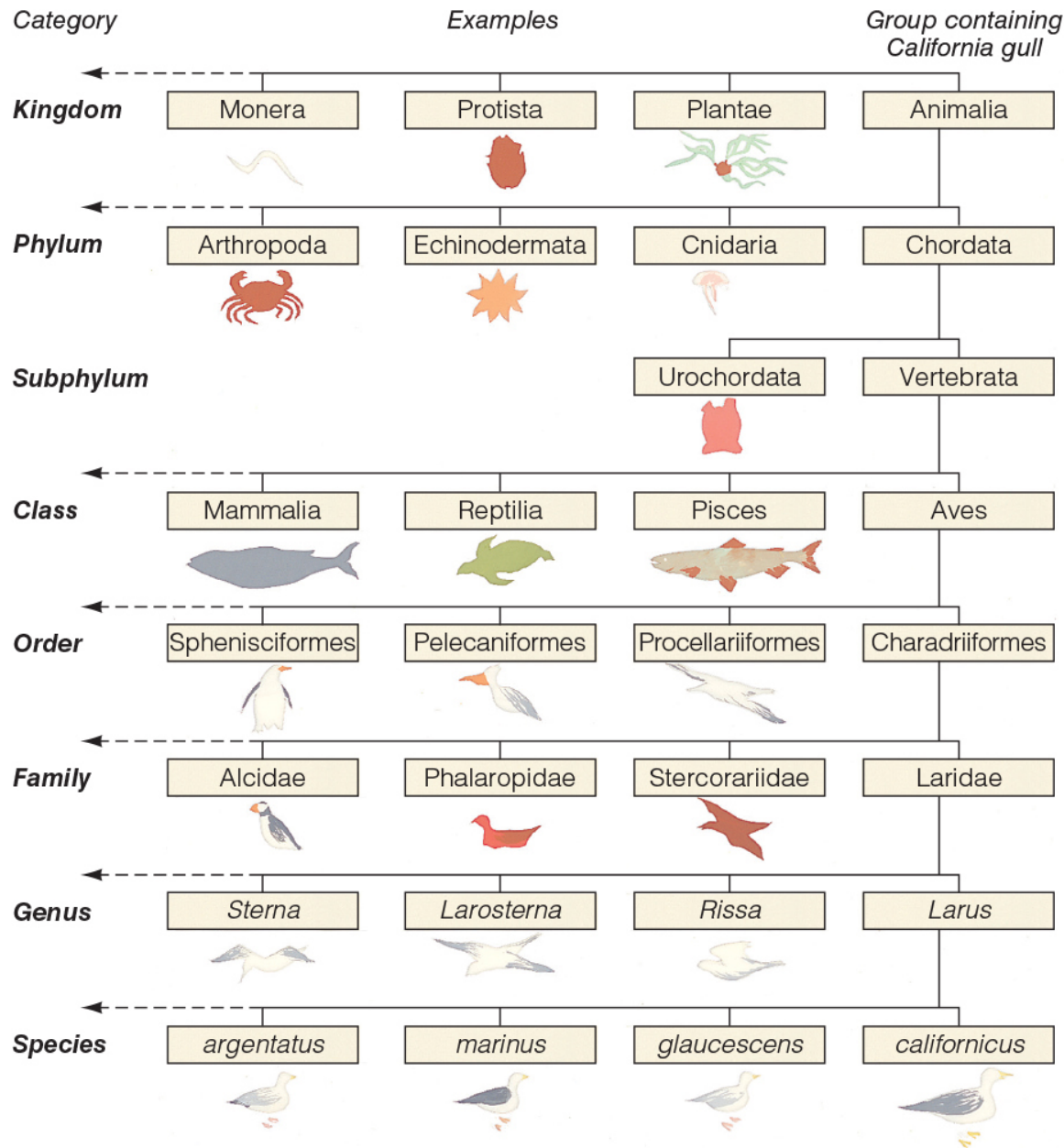


Figure 13.7 A family tree showing the relationship of the three domains of living things presumably evolved from a distant common ancestor. The Bacteria and Archaea contain single-celled organisms without nuclei or organelles; collectively, they are called *prokaryotes*. The fungi, protists, animals, and plants contain organisms with cells having nuclei and organelles; collectively, they are called *eukaryotes*.

Example



An aerial photograph of a coral reef system. The top half of the image shows deep blue ocean water. Below this is a vibrant, multi-colored reef edge with shades of green, yellow, and orange. The bottom half of the image shows a vast, flat, light-colored area, likely a lagoon or a sandy beach, with some darker patches. The title "Flow of Energy and Elements" is centered over the reef edge in a large, white, sans-serif font.

Flow of Energy and Elements

Building towards the concept of the food web

Trophic Pyramid

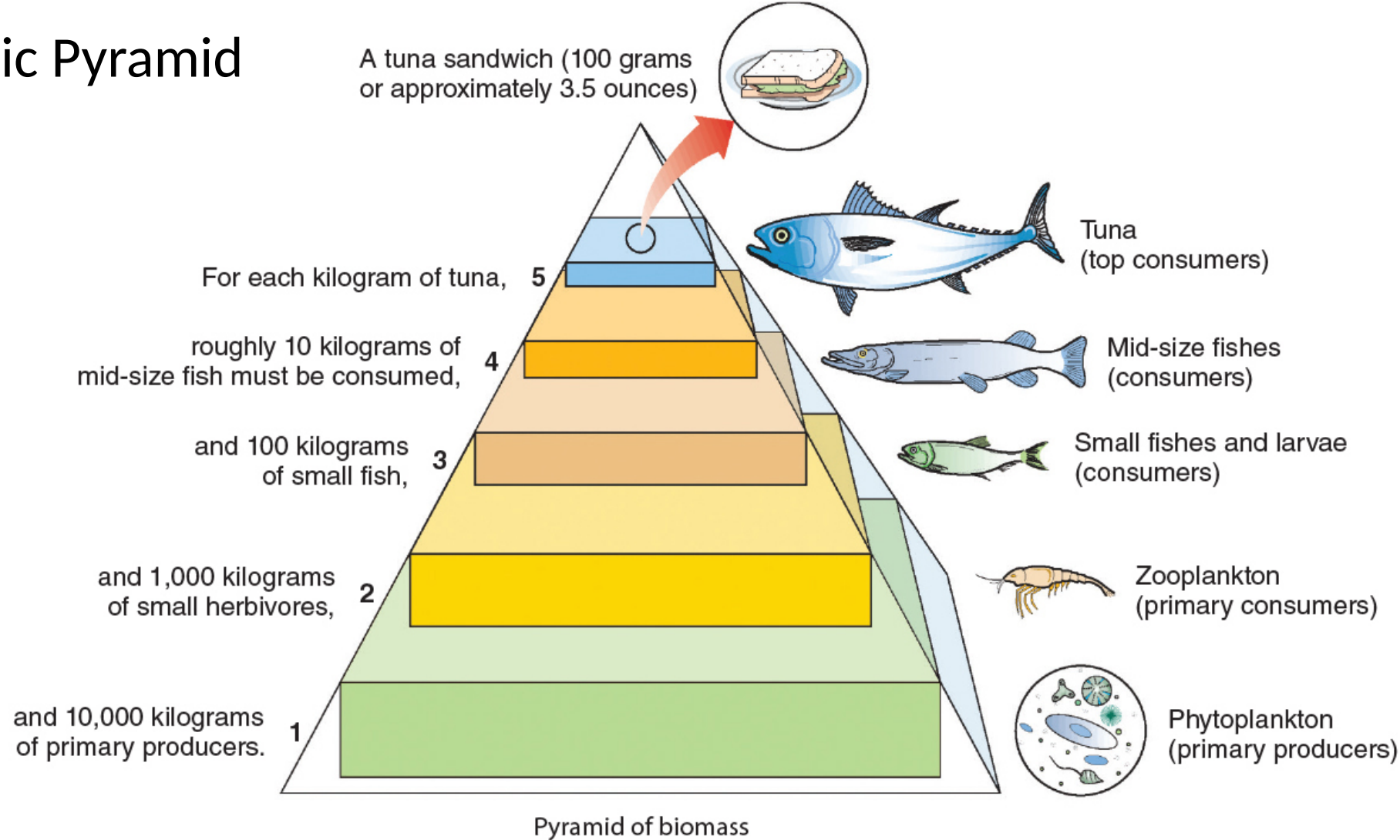


Figure 13.13 A generalized trophic pyramid. How many kilograms of primary producers are necessary to maintain 1 kilogram of tuna, a top carnivore? What is required for an average tuna sandwich? Using the trophic pyramid model shown here, you can see that 1 kilogram of tuna at the fifth trophic level (the fifth feeding step of the pyramid) is supported by 10 kilograms of midsize fish at the fourth, which in turn is supported by 100 kilograms of small fish at the third, which have fed on 1,000 kilograms of zooplankton (primary consumers) at the second, which have eaten 10,000 kilograms of phytoplankton (small autotrophs, primary producers) at the first level. The quarter-pound tuna sandwich has a long and energetic history. (These figures have been rounded off to illustrate the general principle. The actual measurements are difficult to make and quite variable.)

The Marine Food Web

Group of organisms that are linked by complex trophic relationships.

These interactions dictate how energy is flowing through the different trophic levels

In this particular example, everyone depends on sunlight to survive

Autotrophs: make their own food
Heterotrophs: must consume food

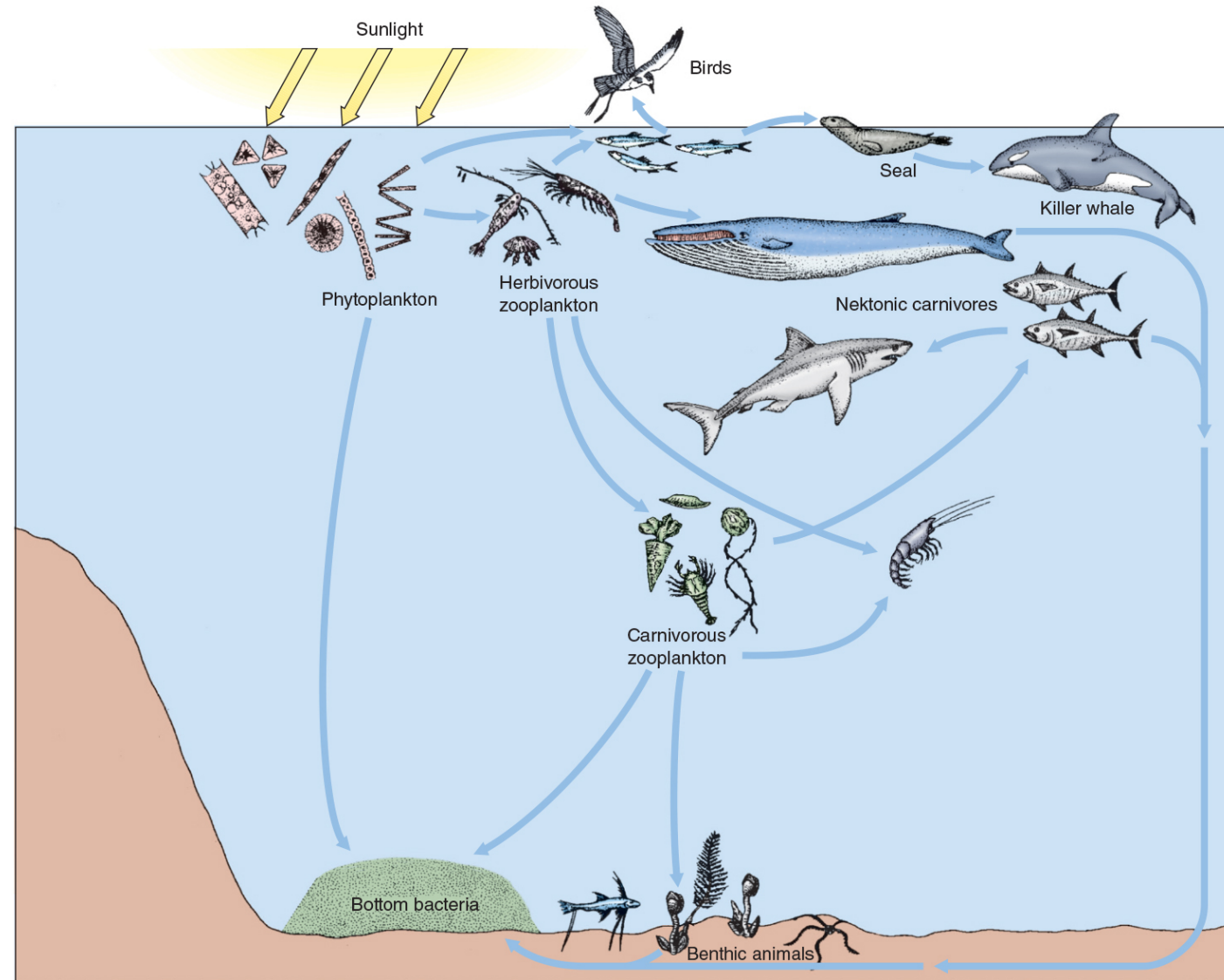
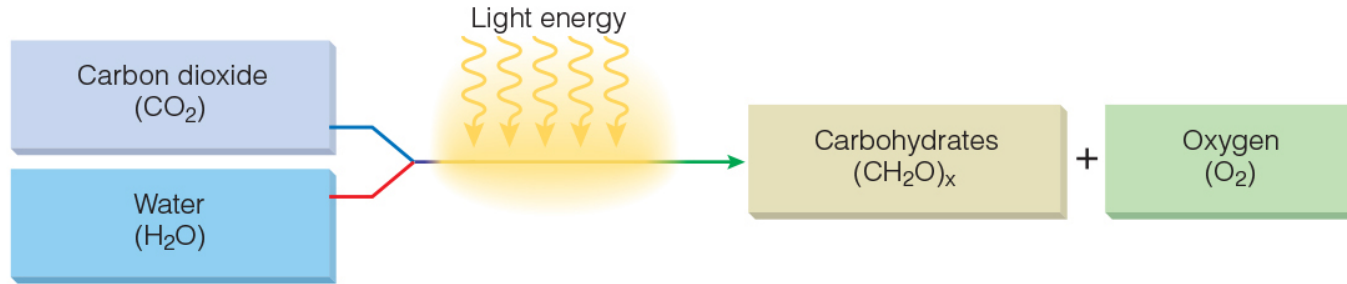


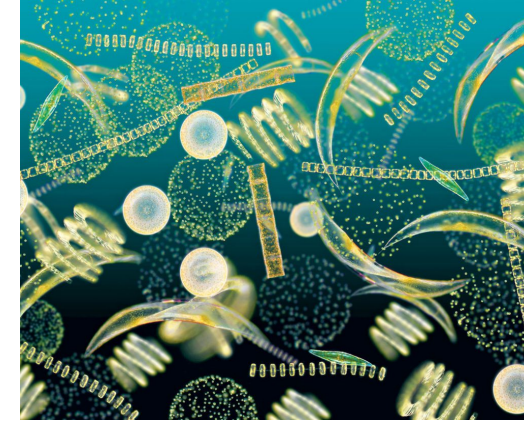
Figure 13.14 A simplified food web, illustrating the major trophic relationships leading to an adult killer whale. The arrows show the direction of energy flow. Note that feeding relationships are not as simple as one might assume from Figure 13.13.

All life depends on Primary Production

Photosynthesis



- a** In photosynthesis, energy from sunlight is used to bond six separate carbon atoms (derived from carbon dioxide) into a single, energy-rich, six-carbon molecule—the sugar glucose—here represented as (CH₂O)_x. The pigment chlorophyll absorbs and briefly stores the light energy needed to drive the reactions. Water is broken down in the process, and oxygen is released.



90-96% of PP

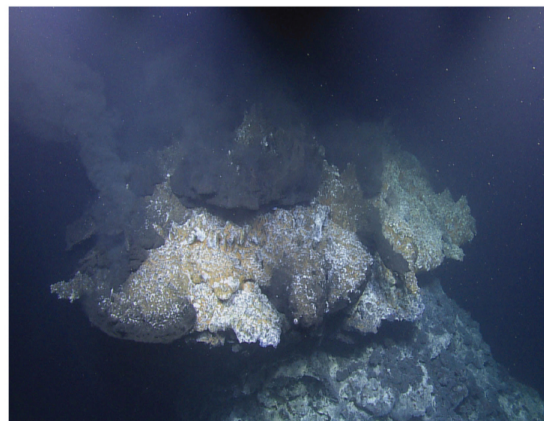
2-5% of PP



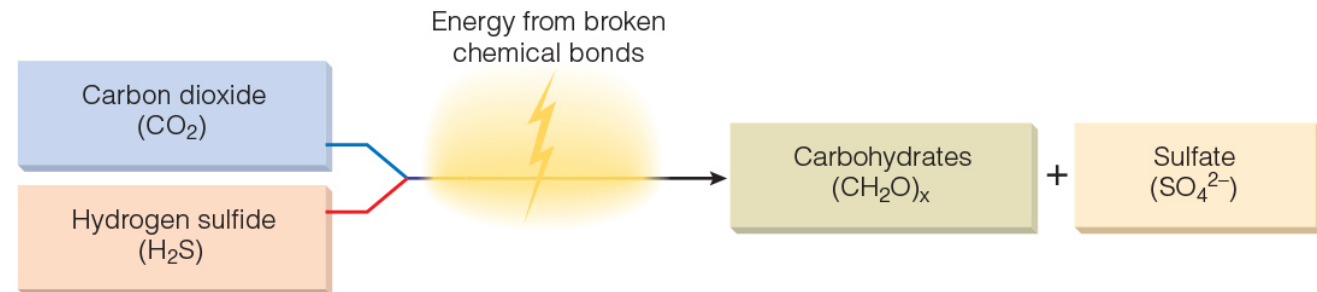
- a** A mass of floating sargassum serves as a food source and habitat for a variety of organisms.

Chemosynthesis

< 5% of PP



- b** White sulfur-oxidizing bacteria blankets the rock of this hydrothermal vent chimney.



- b** A form of chemosynthesis. In this example, 6 molecules of carbon dioxide combine with 6 molecules of oxygen and 24 molecules of hydrogen sulfide to form glucose—(CH₂O)_x. (Other products include molecules of sulfate and water.) The energy to bond carbon atoms into glucose comes from breaking the chemical bonds that hold the sulfur and hydrogen atoms together in hydrogen sulfide.

PHYTOPLANKTON ARE TINY BUT
POWERFUL

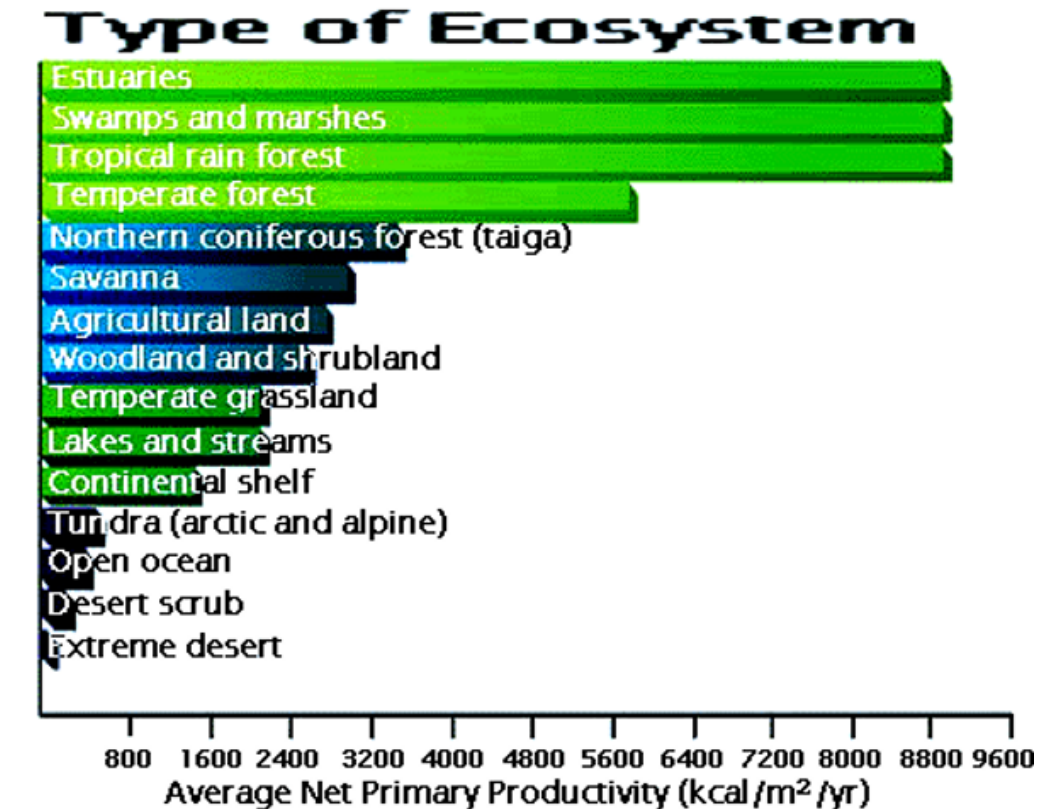


PHYTOPLANKTON ARE TINY BUT POWERFUL

- 50% of global oxygen production
- Relevant for Global Primary Production:
Production of organic matter?

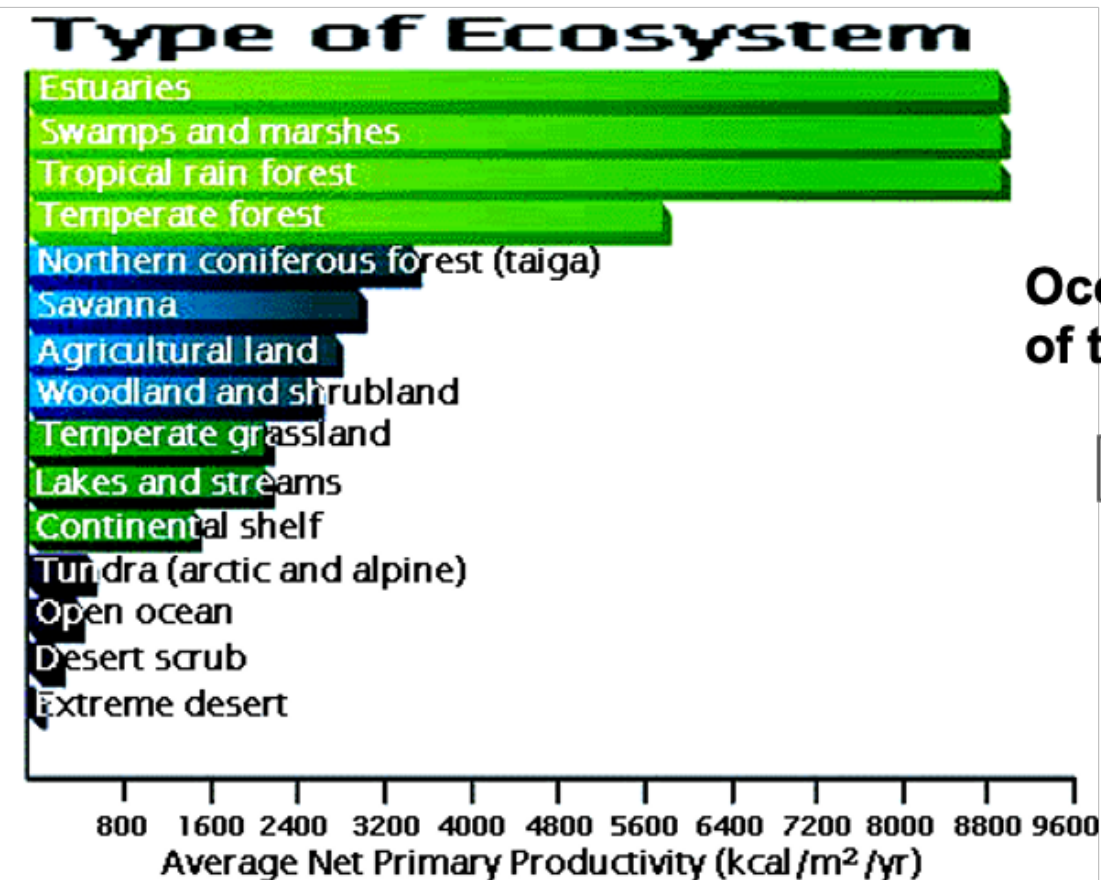
Phytoplankton are the primary producers of the open-oceans, how can they be globally relevant?

A few facts and numbers

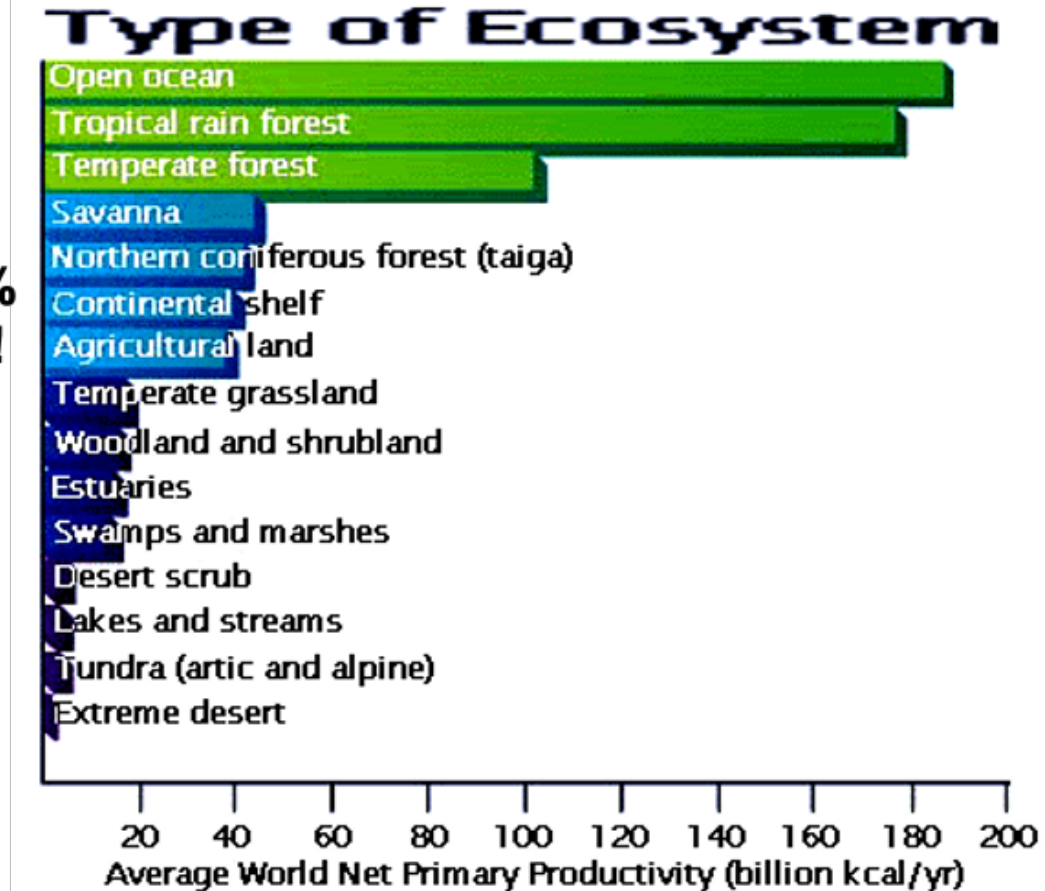


PHYTOPLANKTON ARE TINY BUT POWERFUL

A few facts and numbers



**Ocean occupies 71%
of the Earth surface!**



Open Ocean ~ 26%, Rain Forest ~ 24%

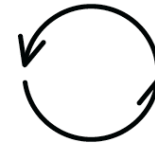
ENVIRONMENTAL FACTORS INFLUENCE THE SUCCESS OF MARINE ORGANISMS



Light



Temperature



Dissolved nutrients



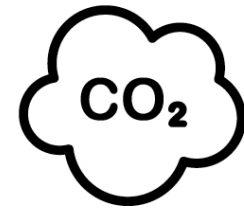
Dissolved gas



Salinity



Hydrostatic pressure

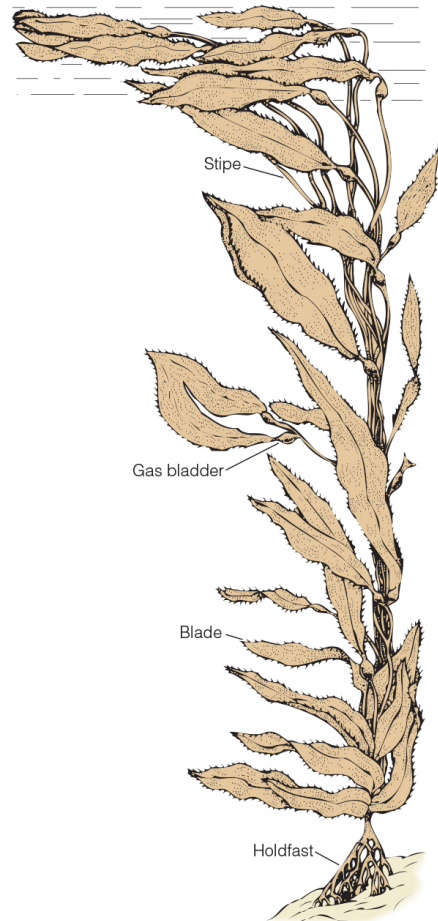


Dissolved acid-base
balance



Note on the big marine plants

- Extremely diverse: species and environments
- Sustain and shape entire ecosystems: coastal and offshore
- More relevant to PP in the coastal ocean
- Only responsible for a minor portion of the marine PP (2-5%)



b *Macrocystis*, a fast-growing, very productive brown alga (phaeophyte), can form spectacular kelp forests off the coast of western North America. Gas bladders at the base of each blade help this alga create a dense canopy at the surface.

a The thallus (body) of a typical multicellular alga. Some of these organisms can grow at a rate of 50 centimeters (20 inches) per day and reach a length of 40 meters (132 feet). Other seaweeds remain small or may simply encrust onto a rocky reef.

Figure 14.16 Seaweeds.



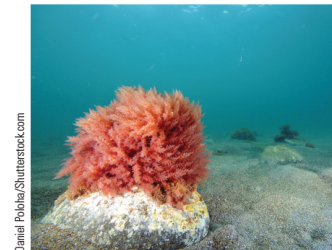
Figure 14.18 Seagrasses are important habitats that provide shelter for many small species and a substrate for other organisms to grow on. While some can live along wave-battered rocky shores, other species live further offshore where they can trap sediments and improve water quality by absorbing nutrients from runoff.



Figure 14.19 A mangrove (*Rhizophora*) growing in brackish water in Belize. The tangled roots descending from the main branches are called *prop roots* or *silt roots*. They provide anchorage for the mangrove, trap sediments, and provide protection for small organisms. Mangroves are also pictured in Figure 12.31.



a Green algae have the same photosynthetic pigments that are found in vascular plants and are also common in freshwater environments.



b Red algae are the most diverse of the seaweeds. They get their characteristic color from the pigment phycoerythrin which strongly absorbs blue light and allows some red algae to live at relatively great depths in comparison to other algae phyla.



c Brown algae are often the most conspicuous of the seaweeds because they include common kelps, rockweeds, and sargassum. They exhibit a great diversity of forms from microscopic filaments to giant kelp and are characterized by the yellowish-brown pigment fucoxanthin that can mask the green chlorophyll.

Figure 14.17 Seaweeds, or multicellular algae, are made up of three different phyla.